



JIMMA UNIVERSITY
JIMMA INSTITUTE OF TECHNOLOGY
DEPARTMENT OF BIOMEDICAL ENGINEERING
BIOMEDICAL IMAGING STREAM

Biomedical artificial intelligence and expert system
Project

Submitted By: Salih Aliyi

Submission to: Dr. Million.M
Submission date: April 2, 2019

Contents	Page
Table of contents.....	II
Introduction.....	1
Expert system.....	1
Representing using the first order logic.....	1
Symptoms.....	2
Treatments.....	5
Prolog terms that are used for this project	6
Flow of the code and Deducing steps.....	7
Over view Results (execution) of Expert System.....	9
Prolog code.....	12
conclusion.....	20

Introduction

This medical expert system is used to diagnose and suggest treatment for the top six (6) respiratory system illness among the patients. The diagnosis is made considering the symptoms that can be seen or felt. This medical expert system helps the doctor or expert in making the appropriate diagnosis of the patient and in prescription of drug and other treatment. the top respiratory system illnesses are:

- Influenzas
- Pneumonia
- Pleural effusion
- Asthma
- Emphysema
- Cystic fibrosis

Expert system

An expert system is a set of programs that manipulate knowledge to solve problems and emulates the decision-making ability of the human expert.

Goal: To design an expert system that can diagnoses Respiratory system illness by verifying the symptoms from the user and suggest treatment.

General description: This expert system predicts the Respiratory system illness types by analyzing the symptoms of the patients and return the type of disease and give how patient can treat himself to overcome the disease.

Representing using the first order logic

First-Order Logic (FOL) is used to represent any kind of situation that are expressed in natural language. And represent objects and relations between objects, variables, and quantifiers.

Expert system for diagnosis of Respiratory system illness and give treatment based on symptoms of the patient we can represented as follows:

symptom

The following symptoms represents **Influenza**:

- ✓ cough
- ✓ fever
- ✓ fatigue
- ✓ runny nose
- ✓ Sore throat
- ✓ Nasal congestion
- ✓ Redness of eyes and skin (specially face)
- ✓ Irritated watering eyes
- ✓ Feeling tired

FOL representation for **Influenza** as the appearing of all this symptoms is must to say the patient have this disease I connect using “AND “ connective and it infer the treatment if all premises are true.

$\text{checksymptom}(\text{cough}) \wedge \text{checksymptom}(\text{fever}) \wedge \text{checksymptom}(\text{fatigue}) \wedge \text{checksymptom}(\text{runny_nose}) \wedge \text{checksymptom}(\text{sore_throat}) \wedge \text{checksymptom}(\text{nasal_congestion}) \wedge \text{checksymptom}(\text{redness_of_eyes_and_skin}(\text{specially_face})) \wedge \text{checksymptom}(\text{irritated_watering_eyes}) \wedge \text{checksymptom}(\text{feeling_tired}) \wedge \text{treatment1} \rightarrow \text{influenza}$

The following symptoms represents **pneumonia**:

- ✓ cough
- ✓ fever
- ✓ fatigue
- ✓ shortness of breath
- ✓ Chest pain during deep breath
- ✓ Blue fringed skin
- ✓ Decrease thirst
- ✓ confusion (decrease level of consciousness)

FOL representation for **pneumonia** as the appearing of all this symptoms is must to say the patient have this disease I connect using “AND “ connective and it infer the treatment if all premises are true.

checksymptom(cough)^checksymptom(fever)^checksymptom(fatigue)^checksymptom(shortness_of_breath)^checksymptom(chest_pain_during_deep_breath)^checksymptom(blue_fringed_skin)^checksymptom(decrease_thirst)^checksymptom(confusion(decrease_level_of_consciousness))^treatment2→ pneumonia

The following symptoms represents **pleural effusion**:

- ✓ Cough
- ✓ fever
- ✓ Shortness of breath
- ✓ Rapid and shallow breathing
- ✓ Pain and swelling in the joint
- ✓ Ventricular tachycardia

FOL representation for **pleural effusion** as the appearing of all this symptoms is must to say the patient have this disease I connect using “AND “ connective and it infer the treatment if all premises are true

checksymptom(cough)^checksymptom(shortness_of_breath)^checksymptom(fever)^checksymptom(rapid_and_shallow_breathing)^checksymptom(pain_and_swelling_in_the_joint)^checksymptom(ventricular_tachy_cardia)^treatment6→pleural effusion

The following symptoms represents **Asthma**:

- ✓ cough
- ✓ shortness of breath
- ✓ Chest tightness
- ✓ wheezing
- ✓ Hard to bring sputum up

Biomedical artificial intelligence and expert system Project

FOL representation for **Asthma** as the appearing of all this symptoms is must to say the patient have this disease I connect using “AND “ connective and it infer the treatment if all premises are true

$\text{checksymptom}(\text{cough}) \wedge \text{checksymptom}(\text{shortness_of_breath}) \wedge \text{checksymptom}(\text{chest_tightness}) \wedge \text{checksymptom}(\text{wheezing}) \wedge \text{checksymptom}(\text{hard_to_bring_sputum_up}) \wedge \text{treatment3} \rightarrow \text{asthma}$

The following symptoms represents **Emphysema**:

- ✓ cough
- ✓ shortness of breath
- ✓ sputum production
- ✓ Easy to bring sputum up
- ✓ Over age of 35

FOL representation for **Emphysema** as the appearing of all this symptoms is must to say the patient have this disease I connect using “AND “ connective and it infer the treatment if all premises are true

$\text{checksymptom}(\text{cough}) \wedge \text{checksymptom}(\text{shortness_of_breath}) \wedge \text{checksymptom}(\text{sputum_production}) \wedge \text{checksymptom}(\text{easy_to_bring_sputum_up}) \wedge \text{checksymptom}(\text{over_age_of_35}) \wedge \text{treatment4} \rightarrow \text{emphysema}$

The following symptoms represents **Cystic fibrosis**:

- ✓ cough
- ✓ shortness of breath
- ✓ Accumulation of thick sticky mucus
- ✓ Salty tasting skin

FOL representation for **Cystic fibrosis** as the appearing of all this symptoms is must to say the patient have this disease I connect using “AND “ connective and it infer the treatment if all premises are true

checksymptom(cough)^checksymptom(shortness_of_breath)^checksymptom(accumulation_of_t
hick_sticky_mucus)^checksymptom(salty_tasting_skin)^ treatment5→cystic fibrosis

Treatments

The following treatments are for **Influenza**:

- ✓ Throat lozenge
- ✓ Get lots of rest
- ✓ Anti-inflammatory drug
- ✓ Anal Getic
- ✓ anti-viral drug
- ✓ cough medicine

The following treatments are for **Pneumonia**:

- ✓ Drink plenty of fluids to help loosen secretion and bring up phlegm
- ✓ Get lots of rest
- ✓ control your fever with aspirin or acetaminophen

The following treatments are for **Asthma**:

Quick medication includes:

- ✓ Albuterol
- ✓ Lev albuterol

others are:

- ✓ praxetorium bromide
- ✓ short acting beta adenine antagonists (SABA)
- ✓ levalbuterol

The following treatments are for **Emphysema**:

Quick medication includes:

- ✓ Long term antibiotics
- ✓ erythromycin

The following treatments are for **Cystic fibrosis**:

- ✓ Use multivitamins (particularly fat-soluble vitamins)
- ✓ bronchi dilators
- ✓ Anti-inflammatory agent
- ✓ pancreatic enzyme supplements
- ✓ mucolytics

The following treatments are for **Pleural effusion**:

- ✓ Use relief of symptoms
- ✓ Indocin
- ✓ Removal of the fluids, air, and blood from the pleural space
- ✓ Rexall
- ✓ Corticated (if you have also TB)

Prolog terms that are used for this project

- i. **write ()**: - used to write a single term to the terminal.
For example: write(a), or write('How are you?')
- ii. **read(X)**: -Reads a term from the keyboard and instantiates variable X to the value of the read term.
This term to be read has to be followed by a dot "." and a white space character (such as an enter or space).
For example, in my project:
 - ✓ to read patient name:
hello:- write('What is your name ?'), **read(X)**, write('#### Hello '), tab(1), write(X), write(' ####').
 - ✓ To reply for symptom:
askQuestion(Question):- **read(Reply)**,((Reply==yes; Reply==y)
-> assert(yes(Question));
assert(no(Question)),fail,nl).

- iii. **assert(X)**:-Adds a new fact or clause to the database. Term is asserted as the last fact or clause with the same key predicate.
For instance:
askQuestion(Question):- read(Reply),((Reply==yes; Reply==y)
-> **assert**(yes(Question));
assert(no(Question)),fail,nl).
- iv. **retract(X)**:-Removes fact or clause X from the database.
Example: undo:-**retract**(yes(_)),fail.
undo:-**retract**(no(_)),fail.
- v. **Fail**:- A predicate that always returns false (i.e. it always fails, causing the whole clause to fail and Prolog to try another branch in the recursion tree.
- vi. **tab(X)**:-Writes an X number of spaces to the terminal.
- vii. **true**:-Always succeed. The predicate true/0 is translated into a single virtual machine instruction.
- viii. nl :-Write a newline character to the current output stream.

Flow of the code and Deducing steps

1. First of all, start the prolog by typing “hello_expert” and it ask you patient name when we reply for that it returns this:
?- hello_expert.
GREETING!!!
WELCOME TO EXPERT SYSTEM FOR DIAGNOSIS OF RESPIRATORY
ILLNESSES AND SUGGEST THEIR TREATMENT!!!
What is your name ?
|: salih_aliyi.
Hello salih_aliyi #####
start by typing "check." to start diagnosis
2. by typing “check.” It start diagnosing.
check:- checkfor(D) ,this make the prolog to check the disease step by step.

3. Checkfor(D):-disease name.

Example. checkfor(influenza):-influenza.

- Influenza have also it's own rules(symptoms) then it jump to check symptoms by the next rule

4. disease name:-checksymptom(a),checksymptom(b),,,,,,,,,,.

Here the program jump to the “check symptom(S)”.

✓ first it checks whether the symptom assigned to “yes” or “no” before:

- when it's already assigned with “yes” it jumps to check the **next symptom for the same disease**, and if the symptom that it checking for it is the last symptom the system **infers (deduce)** as the patient have the disease and suggest the treatment also.
- when it's already assigned with “no” the system **infers (deduce)** as the patient have **no** this disease and it jumps to check for the **next disease**.
- But if the symptom is not assigned before it jumps to next step.

✓ if the symptom is not assigned before neither with “yes” nor “no” the system asks the patient for symptom and wait for patient reply:

- When patient reply is “yes” it jumps to check the **next symptom for the same disease**, and if the symptom that it checking for it is the last symptom the system **infers (deduce)** as the patient have the disease and suggest the treatment also.
- When patient reply is “no” the system **infers (deduce)** as the patient have **no** this disease and it jumps to check for the **next disease**.

5. If the collection symptoms of patient did not match with the premises of any disease the system refer the patient to hospital.

Note. Inference rules that mostly I used in my project is **modes ponens, modes Tollens, Hypothetical Syllogism and, and elimination.**

Over view Results(execution) of Expert System

- a. The system returns the type of disease the patient has and suggest its treatment depending on the patient answer for symptoms.

Example: if patient reply “y” or “yes” for all symptoms of influenza then the system return as follows:

1 ?- check.

PLEASE GIVE ANSWER FOR THE QUESTION

check for "INFLUENZA"

Do u have the symptom cough?

write "yes" or "y" if you have the symptom or "no" if you do not have and put “FULLSTOP(.)” at the end

|: y.

I believe you have "INFLUENZA"

TREAT YOUR SELF BY:

Throat lozenge

Get lots of rest

Anti-inflammatory drug

anal genic

anti-viral drug

cough medicine

THANKS FOR USING THE SYSTEM BYE!!!#####

Biomedical artificial intelligence and expert system Project

- b. The expert system does not ask again the symptom which is already asked for diagnosing of the other disease.

For instance: the symptoms “cough, fever, fatigue” are common for INFLUNZA and PNEMONIA, when we reply “y” for three of them and “no” for the runny nose when diagnosis influenza the expert system returns as the patient have NO influenza because patient of influenza must have also runny nose and do not ask the common symptom again to diagnosis of pneumonia. And when the patient “y” for all symptoms of pneumonia Here the result of prolog:

•	•	•
•	•	•
•	•	•

Do u have the symptom runny nose?

write "yes" or "y" if you have the symptom or "no" if you do not have and put "FULLSTOP(.)" at the end

|: no.

you do not have "INFLUENZA" now check for "PNEMONIA"

Do u have the symptom shortness of breath?

write "yes" or "y" if you have the symptom or "no" if you do not have and put "FULLSTOP(.)" at the end

|: y.

•	•	•
•	•	•
•	•	•

I believe you have "PNEMONIA"

TREAT YOUR SELF BY:

Drink plenty of fluids to help loosen secretion and bring up phlegm

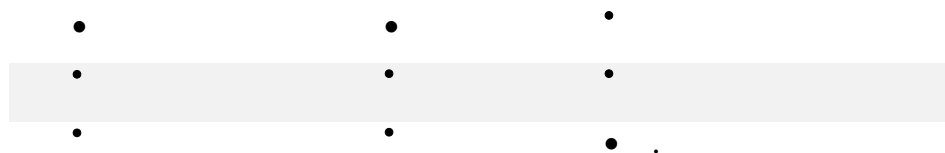
Get lots of rest

control your fever with aspirin or acetal Mino hen

THANKS FOR USING THE SYSTEM BYE!!!#####

- c. If we reply “no” for the symptom which is common for all of disease the expert system returns as it checks all disease that its diagnosis and you are free of the disease and refer the patient to hospital.

For instance when we reply “no” for “cough” :-



Do u have the symptom cough?

write "yes" or "y" if you have the symptom or "no" if you do not have and put "FULLSTOP(.)" at the end

|: no.

you do not have "INFLUENZA" now check for "PNEMONIA"

you do not have "PNEMONIA" also, now check for "PLEURAL EFFUSION"

you do not have "PLEURAL EFFUSION" also, now check for "ASTHMA"

you do not have "ASTHMA" also, now check for "EMPHYSEMA"

you do not have "EMPHYSEMA" also, now check for "CYSTIC FIBROSIS"

you do not have "CYSTIC FIBROSIS"

I CAN NOT IDENTIFY YOUR DISEASE PLEASE YOU BETTER TO GO TO HOSPITAL####

THANKS FOR USING THE SYSTEM BYE!!!#####

HERE IS PROLOG CODE

%%
%%
%%
%%

% JIMMA UNIVERSITY

% JIMMA INSTITUTE OF TECHNOLOGY

% DEPARTEMENT OF BIOMEDICAL ENGINEERING

% ARTEFICAL INTILLIGENCY

%

% PROJECT TITLE: EXPERT SYSTEM FOR DIAGNOSIS RESPIRATORY

% ILLNESSES AND SUGGEST THEIR TREATMENT.

% NAME: SALIH ALIYI

% SUBMITTED TO: DR. MILLION

%

```
Hello_expert:-nl, write('GREETING!!! '),nl,nl,  
  
    write('WELCOME TO EXPERT SYSTEM FOR DIAGNOSIS OF RESPIRATORY  
ILLNESSES AND SUGGEST THEIR TREATMENT!!! '),nl,nl,write('What is your name ?'),nl,nl,  
  
read(X),nl,  
  
write('##### Hello '), tab(1), write(X), write(' #####'),nl,  
  
nl,write('start by typing "check." to start diagnosis'),nl,nl.  
  
check:-  
  
    nl, write('PLEASE GIVE ANSWER FOR THE QUESTION'),nl,nl,  
  
    checkfor(D),nl,bye,nl,nl,  
  
undo.  
  
checkfor(influenza):-nl,write('check for "INFLUENZA"'),nl,  
  
    influenza.  
  
influenza:-  
  
    checksymptom(cough),  
  
    checksymptom(fever),  
  
    checksymptom(fatigue),  
  
    checksymptom(runny_nose),  
  
    checksymptom(sore_throat),
```

Biomedical artificial intelligence and expert system Project

checksymptom(nasal_congestion),

checksymptom(redness_of_eyes_and_skin(specially_face)),

checksymptom(irritated_watering_eyes),

checksymptom(feeling_tired),

treatment1,

nl.

checkfor(pneumonia):-write('you do not have "INFLUENZA" now check for "PNEUMONIA"),nl,

pneumonia.

pneumonia:-

checksymptom(cough),

checksymptom(fever),

checksymptom(fatigue),

checksymptom(shortness_of_breath),

checksymptom(chest_pain_during_deep_breath),

checksymptom(blue_fringed_skin),

checksymptom(decrease_thirst),

checksymptom(confusion(decrease_level_of_consciousness)),

treatment2,

nl.

checkfor(pleural_effusion):-write('you do not have "PNEUMONIA" also, now check for "PLEURAL EFFUSION"),nl,

pleural_effusion.

pleural_effusion:-

checksymptom(cough),

checksymptom(fever),

checksymptom(shortness_of_breath),

checksymptom(rapid_and_shallow_breathing),

checksymptom(pain_and_swelling_in_the_joint),

checksymptom(ventricular_tachy_cardia),

treatment3,

nl.

checkfor(asthma):-write('you donot have "PLEURAL EFFUSION" also, now check for "ASTHMA "'),nl,

asthma.

asthma:-

checksymptom(cough),

checksymptom(shortness_of_breath),

checksymptom(chest_tightness),

checksymptom(wheezing),

checksymptom(hard_to_bring_sputum_up),

treatment4,

nl.

checkfor(emphysema):-write('you donot have "ASTHMA" also, now check for "EMPHYSEMA "'),nl,

emphysema.

emphysema:-

checksymptom(cough),

checksymptom(shortness_of_breath),

Biomedical artificial intelligence and expert system Project

checksymptom(sputum_production),

checksymptom(easy_to_bring_sputum_up),

checksymptom(over_age_of_35),

treatment5,

nl.

checkfor(cystic_fibrosis):-write('you donot have "EMPHYSEMA" also, now check for" CYSTIC FIBROSIS"),nl,

cystic_fibrosis.

cystic_fibrosis:-

checksymptom(cough),

checksymptom(shortness_of_breath),

checksymptom(accumulation_of_thick_sticky_mocus),

checksymptom(salty_tasting_skin),

treatment6;last.

askQuestion(Question):-nl,

write('Do u have the symptom '),

write(Question),

write('?'),

nl,nl,

write('write "yes" or "y" if you have the symptom or "no" if you do not have and put"FULLSTOP(.)" at the end'),

nl,

read(Reply),

```
( (Reply==yes; Reply==y)
```

```
-> assert(yes(Question));
```

```
assert(no(Question)),fail,nl).
```

```
:-dynamic yes/1,no/1.
```

```
checksymptom(S):-
```

```
( yes(S)
```

```
-> true;
```

```
( no(S)
```

```
->nl,fail,nl,nl,nl;
```

```
askQuestion(S))).
```

```
treatment1:-nl,write('I believe you have "INFLUENZA" '),nl
```

```
,write('TREAT YOUR SELF BY: '),
```

```
nl,write('# Throat lozenge'),nl,
```

```
write('# Get lots of rest'),nl
```

```
,write('# Anti inflammatory drug'),nl,
```

```
write('# anal gesic '),nl
```

```
,write('# anti viral drug'),nl
```

```
,write('# cough medicine')
```

```
.
```

```
treatment2:-nl,write('I believe you have "PNEMONIA" '),nl,
```

```
write('TREAT YOUR SELF BY: '),
```

```
nl,
```

```
write('# Drink plenty of fluids to help loosen secretion and bring up phlegm'),nl,
```

Biomedical artificial intelligence and expert system Project

```
write('# Get lots of rest'),nl
,write('# control your fever with aspirin or acetaminophen')
.

treatment3:-nl,write('I believe you have "PNEUMONIAL EFFUSION" '),nl,write('TREAT YOUR
SELF BY: '),
nl,
nl,write('# Use relief of symptoms '),nl,
write('# Indocin'),nl,
write('# Removal of the fluids,air,and blood from the pleural space '),nl,
write('# trexall'),nl,
write('# corticosteroid(if you have also TB)'),nl
.

treatment4:-nl,write('I believe you have "ASTHMA" '),nl,
        write('TREAT YOUR SELF BY: '),
nl,
nl,
        write(' Quick medication include: '),nl,
write('# Albuterol'),nl,
write('# levalbuterol'),nl,
write(' Others are:'),nl,nl,
write('# ipratropium bromide'),nl,
write('# short acting beta adrenergic agonist(SABA)'),nl,write(' - levalbuterol')
.
```

Biomedical artificial intelligence and expert system Project

treatment5:-nl,write('I believe you have "EMPHYSMA" '),nl,write('TREAT YOUR SELF BY: '),
nl,

nl,write(' Quick medication include: '),nl,

write('# Long term anti biotics'),nl,

write('# erythronycin')

.

treatment6:-nl,write('I believe you have "CYSTIC FIBROSIS" '),nl,

write('TREAT YOUR SELF BY: '),

nl,

nl,write('# Use multivitamins(particularly fat soluble vitamins) '),nl,

write('# bronchi dilators'),nl,write('# Anti inflammatory agent'),nl,

write('# pancreatic enzyme supplements'),nl,write('- mucolytics'),nl

.

last:-write('you donot have "CYSTIC FIBROSIS"'),nl,nl,nl,write('####I CAN NOT IDENTIFY
YOUR DISEASE PLEASE YOU BETTER TO GO TO HOSPITAL####'),nl.

bye:-nl,write('##### THANKS FOR USING THE SYSTEM BYE!!!#####'),nl,nl.

undo:-retract(yes(_)),fail.

undo:-retract(no(_)),fail.

undo.

Conclusion

This expert system is used for the diagnosis of six most common respiratory system illnesses and suggest treatment for each of them. and if it doesn't identify the disease that the patient has system tells the patient as it checks for those disease and conclude as the patient have no any of them and refer the patient to the hospital. This expert system can be used by anyone who needs it.